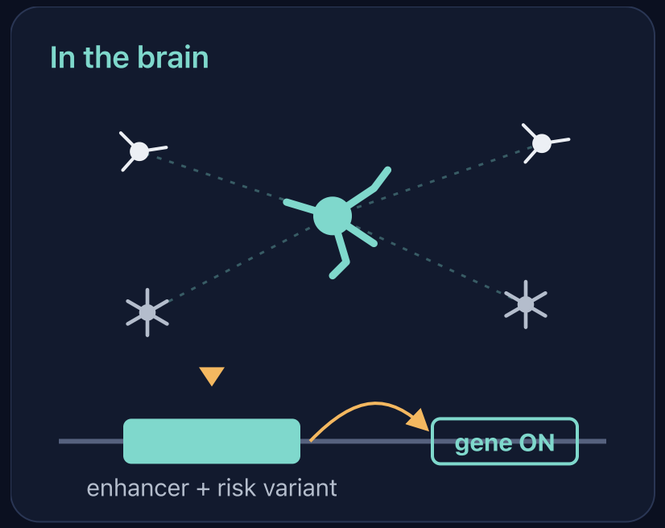
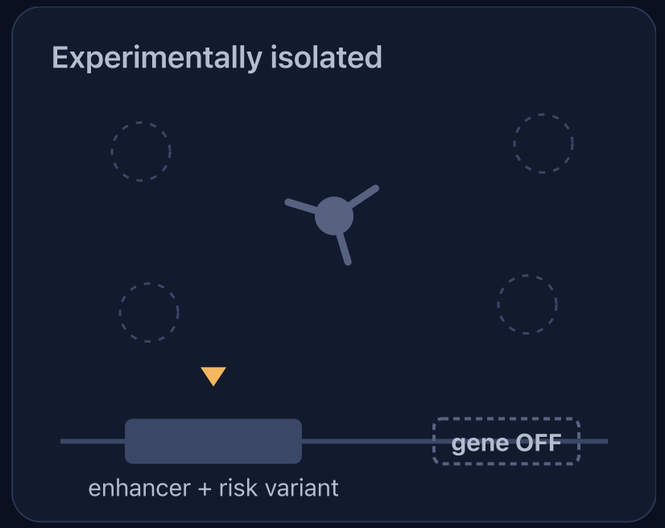
We can screen what a gene does to a cell. We can’t yet screen *what a cell does to a cell.*

Alzheimer’s risk sits in the regulatory DNA of microglia, yet microglia lose that state the moment they are experimentally isolated.



Risk variants: “largely confined to microglia enhancers”
Nott 2019



Disease-linked genes switch off; survival needs astrocyte signals
Gosselin 2017 · Bohlen 2017

152.8M



projected dementia cases by 2050 (57.4M in 2019)

0.4%



of 413 AD trials succeeded (2002–12)

~1 in 3



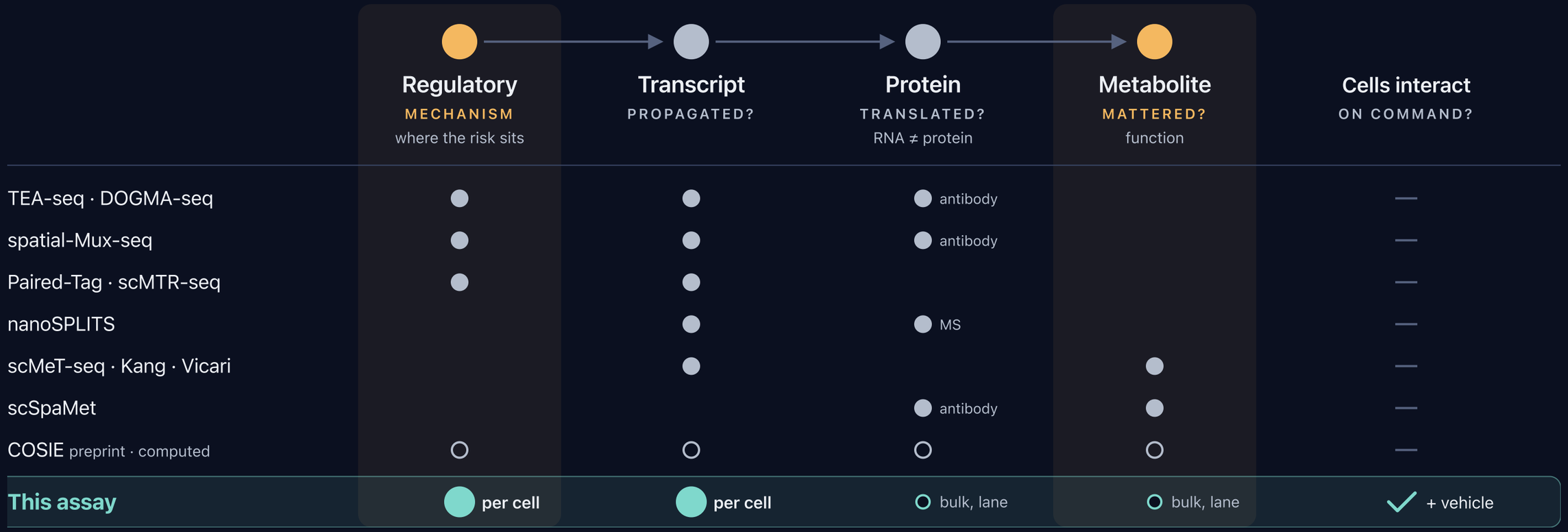
putative AD risk genes lack an adequate mouse ortholog (Mancuso 2019)



Why these three cells: microglia can switch astrocytes into a neurotoxic state.
Liddelow 2017 · a defined human iPSC tri-culture system exists: Guttikonda 2021

Sources | Nott 2019 10.1126/science.aay0793 · Gosselin 2017 10.1126/science.aal3222 · Bohlen 2017 10.1016/j.neuron.2017.04.043 · Mancuso 2019 10.1038/s41593-019-0525-x
GBD 2019 10.1016/S2468-2667(21)00249-8 · Cummings 2014 10.1186/alzrt269 · Liddelow 2017 10.1038/nature21029 · Guttikonda 2021 10.1038/s41593-020-00796-z

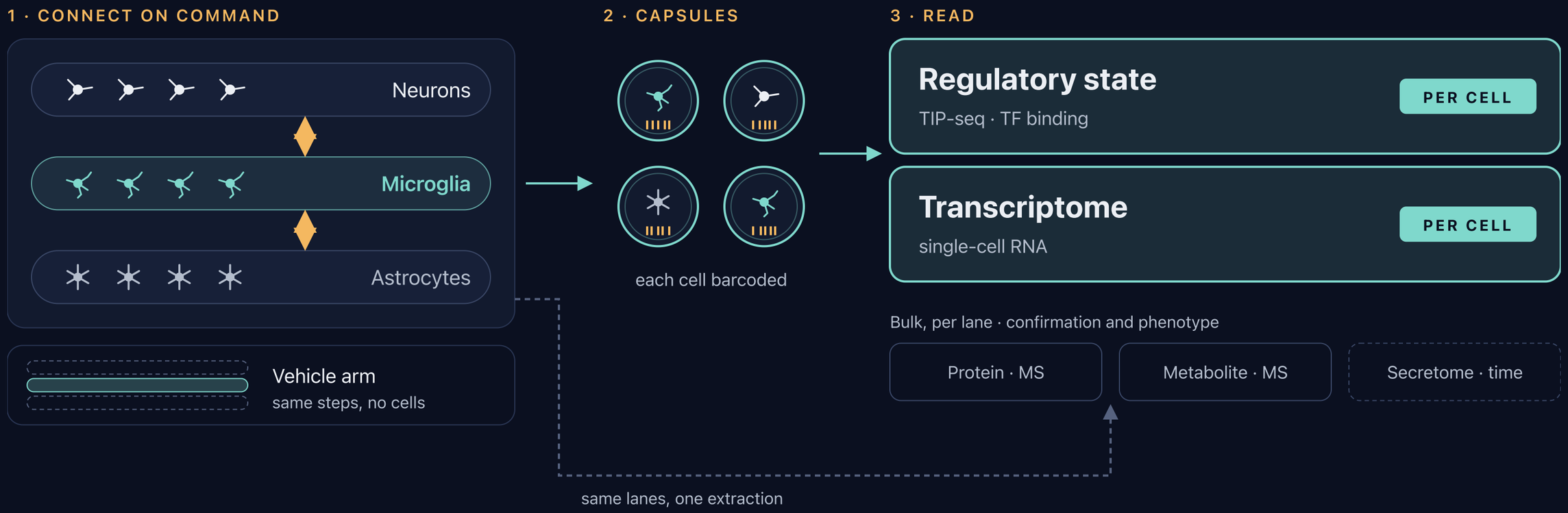
We found no published method that measures *both ends of the chain* in one cell population.



Sources | TEA-seq 10.7554/eLife.63632 · DOGMA-seq 10.1038/s41587-021-00927-2 · spatial-Mux-seq 10.1038/s41592-024-02576-0 · Paired-Tag 10.1038/s41592-021-01060-3 · scMTR-seq 10.1126/sciadv.adu3308 · nanoSPLITS 10.1038/s41467-024-54099-z · scMeT-seq 10.1002/advs.202411276 · Kang (plant) 10.1073/pnas.2512828122 · Vicari 10.1038/s41587-023-01937-y · scSpaMet 10.1038/s41467-023-43917-5 · COSIE 10.64898/2026.03.03.709303 · RNA ≠ protein: Liu 2016 10.1016/j.cell.2016.03.014 · Metabolism as microglial function (Aβ: OXPHOS → glycolysis → tolerant state): Baik 2019 10.1016/j.cmet.2019.06.005. Search not exhaustive.

03 THE ASSAY

Perturb-seq, except the perturbation is *one cell type acting on another.*

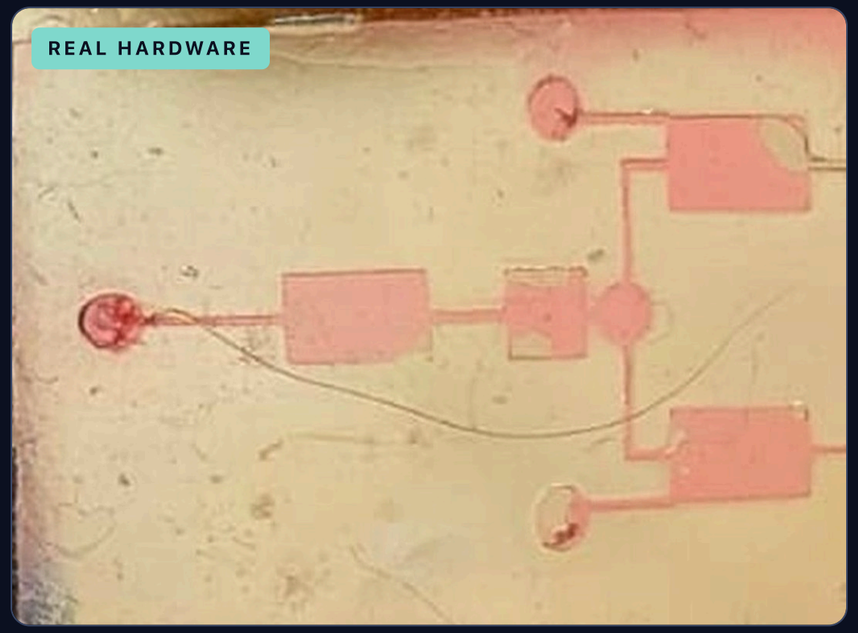


LIMITS Protein and metabolite are lane-level, not single-cell · Demo chip is single-layer and valveless: on-demand valves or contact-only barriers need a different fabrication process · Capsule barcoding of TIP-seq is unpublished

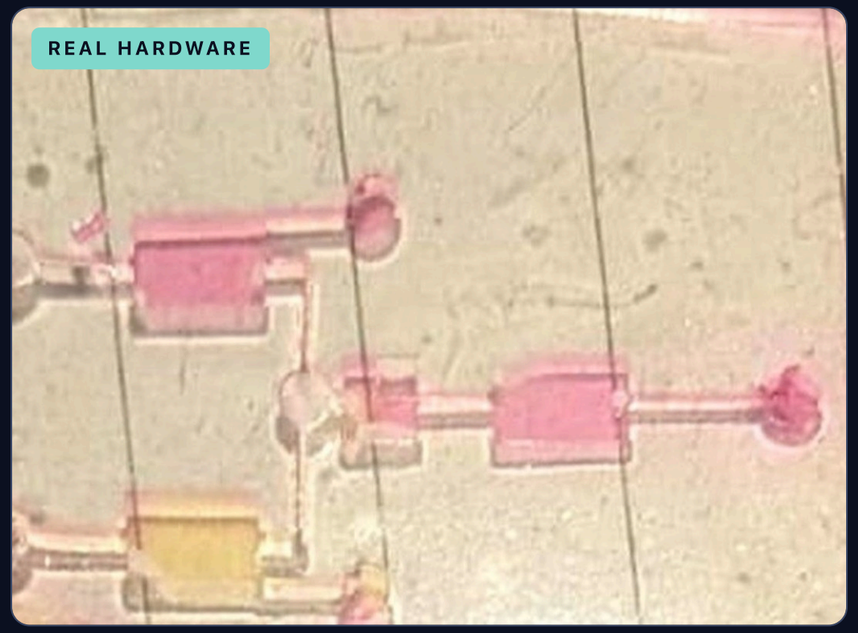
Source | TIP-seq: Bartlett 2021 10.1083/jcb.202103078 (shown for histone marks, RNAPII, CTCF; not yet microglial TFs) · Assay design: project records, unpublished.

A protocol goes in. *A printed chip* comes out.

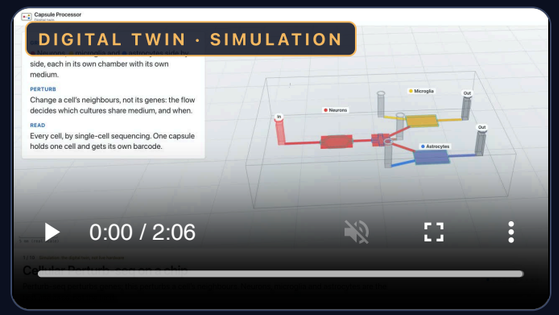
1 Ingest · 2 Extract · 3 Decide · 4 Split · 5 Size · 6 Route · 7 Write · 8 Verify · 9 Simulate · **10 Print**



Module 1, printed and flooded with food dye: dye fills chambers, channels and ports.



Pink in one chamber, yellow in the next, held apart: the grow-apart baseline.



Not live hardware. Every design is simulated first; it caught particle settling in a feed line and a narrow retention-gap tolerance before printing.

STATUS 6 chips printed, single-layer and valveless · food dye, not cells · no bench video yet, only these photographs

Source | Bench photographs, design and simulation records: project, unpublished. No external citation applies.

Small cultures are the point. One run, *about eighty conditions.*



Microlitre volumes: less reagent,
more concentrated sample

FIRST RUN: DOES THE MODEL PICK THE RIGHT SENDER?



6 directed pairs
each chamber takes a turn as sender

3 baselines
grown apart

1 vehicle

One well, 10 × 2,000 cells. Barcode plus chamber label shows which receivers really changed.

3 CELL TYPES IS THE PROOF OF CONCEPT



Now: our use case
3 brain cell types

Next: 10+ compartments
talking to each other; combinations multiply

STATUS 6 dye-only chips · no cells on chip yet · trap needs a bought mesh · 3 steps still at the bench · full scale simulated, not built

Source | Cell numbers per well and per condition: project sizing, unpublished.